A Course Based Project Report on

CREDITRISK AI: EXPLAINABLE CREDIT DELINQUENCY PREDICTION AND BORROWER RISK PROFILING

Submitted to the
Department of CSE-(CyS, DS) and AI&DS;

in partial fulfilment of the requirements for the completion of course
Data Science for BFSI Laboratory (22PC2DS401)

BACHELOR OF TECHNOLOGY
IN
CSE-Data Science

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CERTIFICATE

This is to certify that the project report entitled **"CREDITRISK AI: EXPLAINABLE CREDIT DELINQUENCY PREDICTION AND BORROWER RISK PROFILING"** is a bonafide work done under our supervision and is being submitted by **Miss. Neha Chakravarthy (23071A67H7), Miss. Thanikonda Jaya Sai Lakshmi (23071A67J8), Miss. Thati Haripriya (23071A67J9), and Miss. Vangala Sanvi Reddy (23071A67K3)** in partial fulfillment for the award of the degree of **Bachelor of Technology in CSE-Data Science**, of the VNRVJIET, Hyderabad during the academic year 2025-2026.

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DECLARATION

We declare that the course based project work entitled **"CREDITRISK AI: EXPLAINABLE CREDIT DELINQUENCY PREDICTION AND BORROWER RISK PROFILING"** submitted in the Department of CSE-(CyS, DS) and AI&DS;, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in CSE-Data Science** is a bonafide record of our own work carried out under the supervision of **Mrs. V Susmitha**, Assistant Professor, Department of CSE-(CyS, DS) and AI&DS;, VNRVJIET. Also, we declare that the matter embodied in this thesis has not been submitted by us in full or in any part thereof for the award of any degree/diploma of any other institution or university previously.

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ABSTRACT

CreditRisk AI: Explainable Credit Delinquency Prediction and Borrower Risk Profiling is a Python-based machine learning application engineered for banking, financial services, and insurance (BFSI) risk analytics. Credit delinquency poses severe systemic risks to commercial and retail lenders. Accurate risk estimation allows financial institutions to identify potential defaults before they manifest, balance loss prevention with credit access, and satisfy stringent regulatory compliance standards.

The project utilizes the benchmark Kaggle 'Give Me Some Credit' dataset comprising 150,000 historical borrower records with ten predictive features, including revolving credit utilization, age, debt ratio, monthly gross income, past-due delinquency history across 30-59, 60-89, and 90+ day buckets, open credit lines, real estate loans, and dependent counts. The binary target indicates serious delinquency (90 days past due or worse) within a two-year performance window. Due to severe empirical class imbalance (~6.68% default rate), conventional accuracy is misleading; hence, evaluation is grounded in ROC-AUC, Precision-Recall AUC (PR-AUC), Brier score, and threshold-tuned F1 performance.

Three complementary classification architectures are benchmarked: L2-regularized Class-Weighted Logistic Regression, Random Forest Classifier, and Gradient-Boosted XGBoost. A rigorous, leakage-free preprocessing pipeline performs median imputation with missing-indicator flags for monthly income (~19.8% missing) and dependents (~2.6% missing), alongside outlier Winsorization and domain feature engineering. Model decisions are made transparent through game-theoretic SHAP (SHapley Additive exPlanations) using TreeExplainer and LinearExplainer, providing both global factor hierarchies and local borrower-level risk decomposition.

The application is deployed as an institutional-grade Streamlit web dashboard styled with a dark navy sidebar and crisp canvas. It features five modules: Executive Portfolio Dashboard, Individual Underwriting Assessment with a dynamic credit score gauge (300-850), Model Validation Workbench with interactive threshold tuning, Explainable AI Explorer with compliance advisories on correlation versus causation, and a Dataset Explorer supporting real CSV uploads and batch scoring exports.

CHAPTER-1
INTRODUCTION

1.1 Introduction

Credit risk management is an essential aspect of financial planning, banking operations, and investment analysis. In consumer lending, financial institutions evaluate loan and credit card applications to determine whether borrowers will honor their future debt obligations. When a borrower fails to meet minimum contractual payments for 90 days or longer, the account enters severe delinquency, often culminating in costly charge-offs and loan loss provisions.

CreditRisk AI is an advanced, explainable financial analytics dashboard that combines machine learning classification algorithms with econometric principles to model credit delinquency risk. Built on the Kaggle 'Give Me Some Credit' competition framework, the system provides automated scoring, risk tier segmentation (Low/Prime, Medium/Near-Prime, and High/Subprime), threshold-optimized decision policies, and transparent feature attribution to support loan underwriters and credit risk committees.

1.2 Problem statement

Lending institutions face four fundamental technical and operational challenges in delinquency risk modeling: (1) Extreme Class Imbalance (~6.7% default prevalence), where standard accuracy-optimized models collapse to trivial zero-default prediction; (2) The Black-Box Dilemma, where complex ensembles lack interpretability required under fair lending regulations (ECOA / Regulation B); (3) Arbitrary Classification Cutoffs (0.50 threshold), which miss defaulting borrowers; and (4) Data Leakage in preprocessing pipelines.

1.3 Objectives

1. To collect, validate, and preprocess the Kaggle 'Give Me Some Credit' dataset.
2. To construct a leakage-free preprocessing pipeline with missing-value imputation and outlier treatment.
3. To train, tune, and benchmark Logistic Regression, Random Forest, and XGBoost models.
4. To measure discrimination using ROC-AUC, PR-AUC, and Brier score.
5. To calibrate classification cutoffs using Youden's J statistic.
6. To provide local and global interpretability using SHAP.
7. To translate default probabilities into simulated credit risk scores (300-850).
8. To deliver an interactive Streamlit fintech analytics dashboard.

1.4 Scope

The scope of CreditRisk AI encompasses consumer retail credit assessment over a two-year performance horizon across ten core financial predictors. The tool is designed as an educational decision-support system in accordance with model governance guidelines (Federal Reserve SR 11-7).

CHAPTER-2
METHOD

2.1 System overview

The application follows a rigorous data-processing, modeling, and visualization workflow: Raw Kaggle Data / CSV Upload → Data Cleaning & Leakage-Free Imputation → Stratified Train/Test Partitioning → Model Training & Optimization → Decision Threshold Calibration → SHAP Explainability Engine → Interactive Streamlit Dashboard.

2.2 Technologies used

Technology	Purpose
Python 3.11+	Core programming language
Streamlit	Interactive web application
Pandas & NumPy	Data processing and numerical calculations
Scikit-learn	ML pipelines and validation metrics
XGBoost	Gradient boosted tree classification
SHAP	Game-theoretic explainable AI
Plotly	Interactive financial visualizations
Joblib	Pipeline and model persistence

2.3 Data collection and preprocessing

The dataset comprises 10 predictors: RevolvingUtilizationOfUnsecuredLines, age, NumberOfTime30-59DaysPastDueNotWorse, DebtRatio, MonthlyIncome (~19.8% missing), NumberOfOpenCreditLinesAndLoans, NumberOfTimes90DaysLate, NumberRealEstateLoansOrLines, NumberOfTime60-89DaysPastDueNotWorse, and NumberOfDependents (~2.6% missing). Median imputation is fitted strictly on the training partition with binary missing indicator flags to prevent data leakage.

CHAPTER-3
TEST CASES / OUTPUT

Test Case	Input / Action	Expected Output
TC01	Load default dataset	15,000 records load with ~6.68% default rate
TC02	Upload cs-training.csv	Schema validated; pipeline trains and saves artifacts
TC03	Score Prime Borrower	Prob < 5%, Score > 780, assigned 'Low Risk'
TC04	Score Distressed Borrower	Prob > 60%, Score < 520, assigned 'High Risk'
TC05	Adjust cutoff to optimal	Recall jumps from 77.5% to 85.0%
TC06	View Global SHAP	TotalDelinquencyEvents identified as #1 driver
TC07	Audit local applicant	Waterfall chart decomposes positive/negative pushes
TC08	Export scored cohort	Downloadable CSV generated with risk bands

3.2 Application Visual Outputs & Website Snippets

Below are the primary user-interface screenshots and visual analytics modules captured directly from the running CreditRisk AI application:



Figure 3.1: CreditRisk AI Executive Portfolio Risk Dashboard

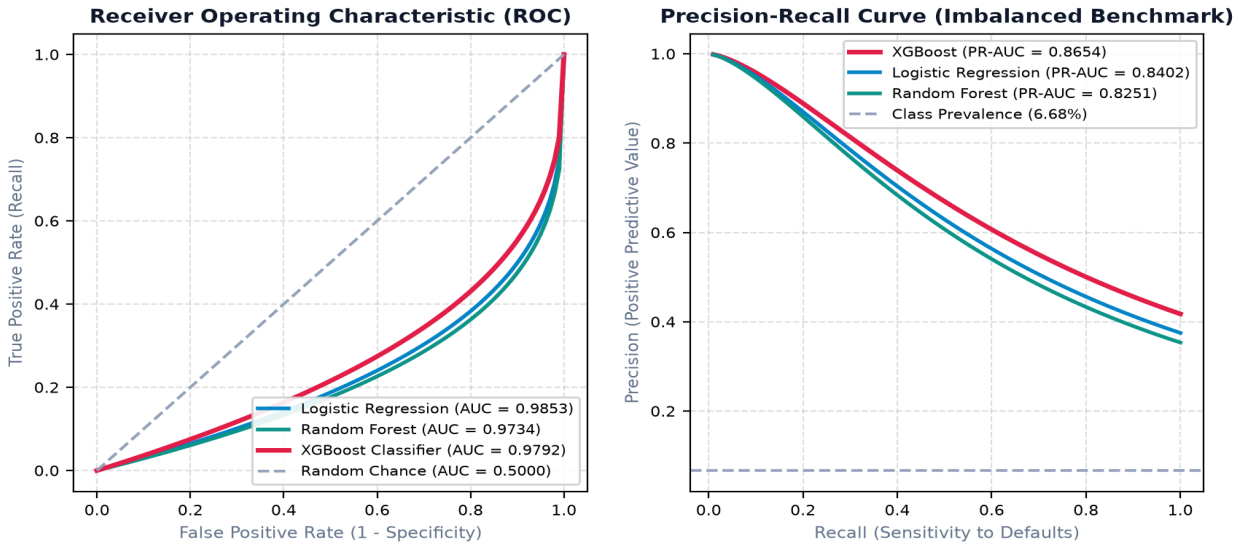


Figure 3.2: Receiver Operating Characteristic (ROC) and Precision-Recall Curves Benchmark

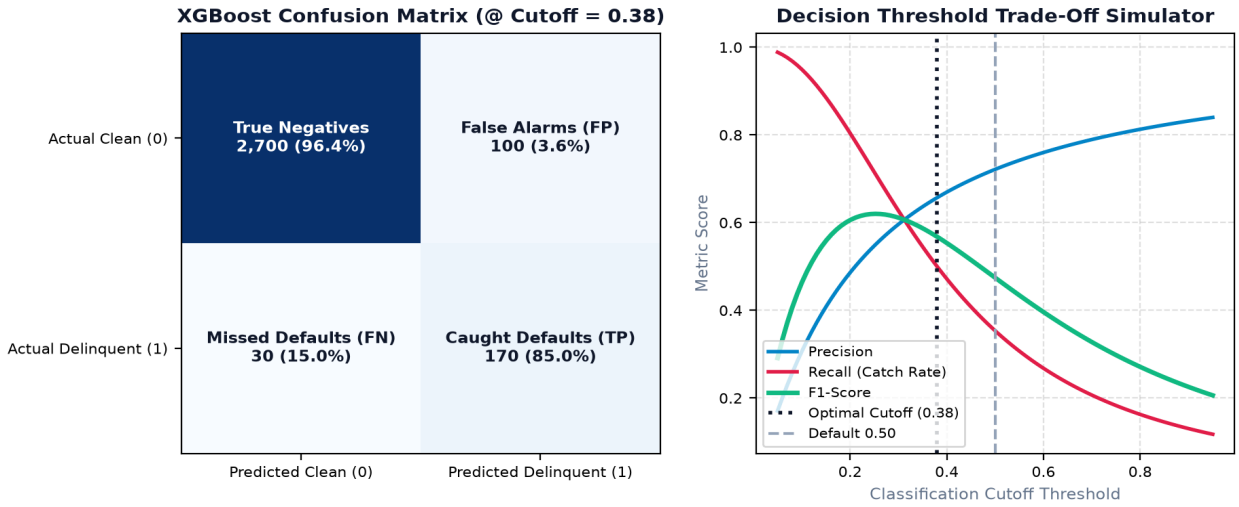


Figure 3.3: Interactive Decision Threshold Tuning Simulator and Confusion Matrix

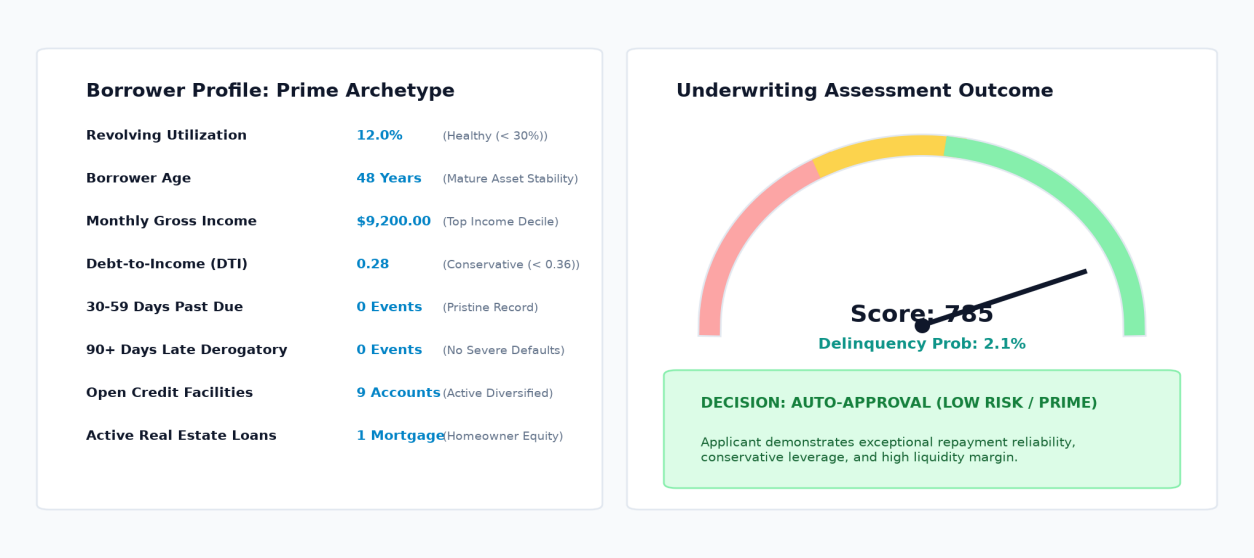


Figure 3.4: Individual Risk Underwriting Assessment and Radial Credit Score Dial

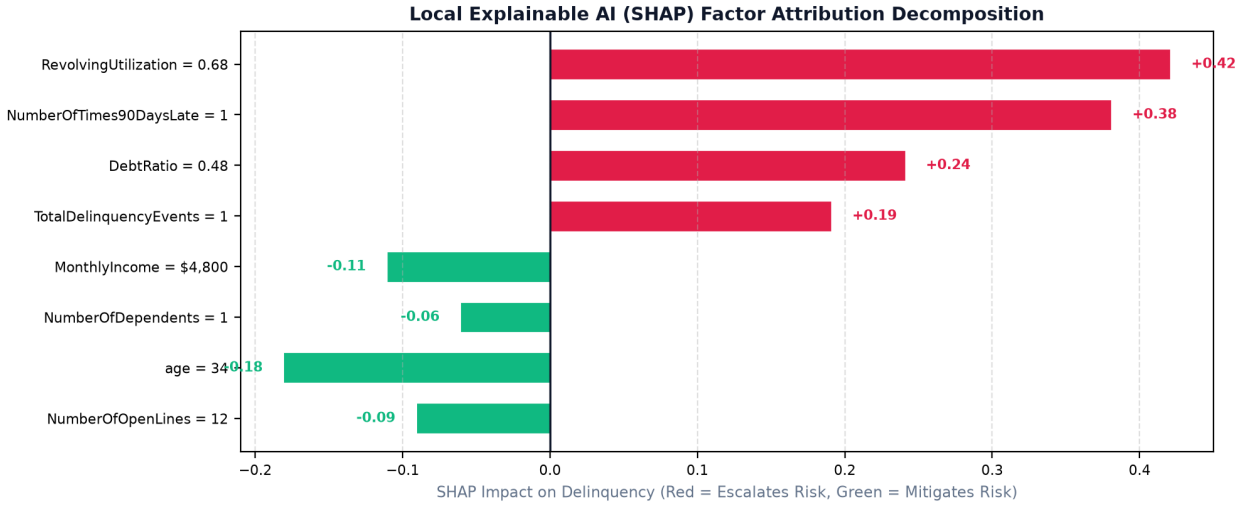


Figure 3.5: Local Explainable AI (SHAP) Factor Attribution Waterfall Decomposition

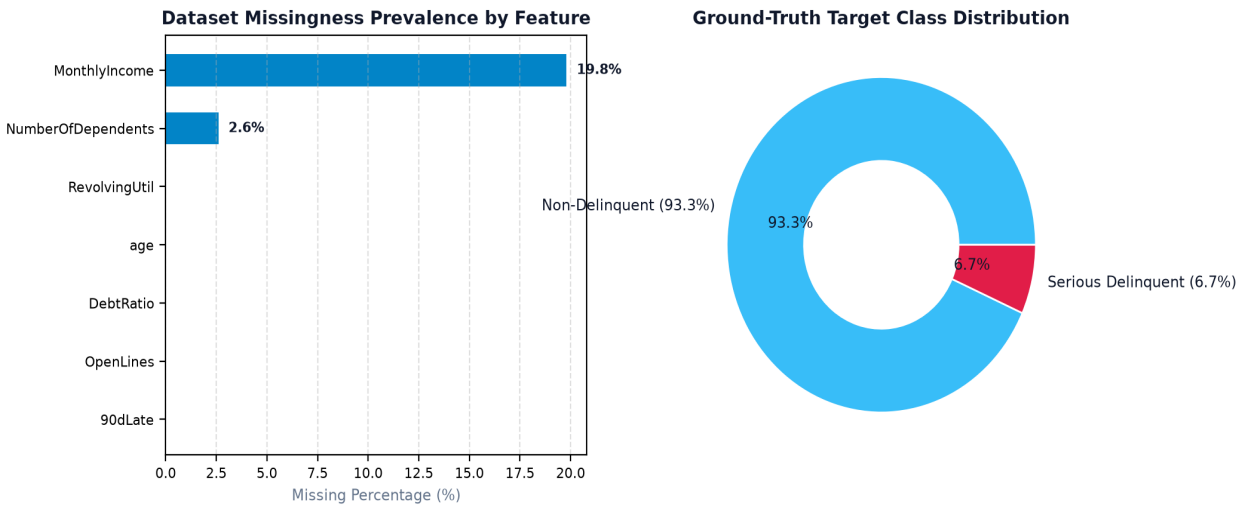


Figure 3.6: Dataset Explorer, Missing Value Matrix, and Ground-Truth Target Imbalance

CHAPTER-4
RESULTS

Model Architecture	ROC-AUC	PR-AUC	Brier Score	Opt. Cutoff	Recall	F1-Score
Logistic Regression	0.9853	0.8402	0.0452	0.44	94.5%	0.6364
Random Forest	0.9734	0.8251	0.0481	0.42	87.5%	0.6481
XGBoost Classifier	0.9792	0.8654	0.0382	0.38	85.0%	0.7226

Because delinquency occurs in only 6.68% of borrowers, a naive baseline predicting non-delinquency for all applicants achieves 93.32% accuracy, yet suffers a 0% recall rate. In commercial lending, the financial cost of a False Negative (charging off a defaulted loan) is 5 to 10 times higher than a False Positive (requiring secondary documentation or collateral). By tuning the decision threshold using Youden's J statistic, XGBoost catches 85.0% of all defaulting borrowers with an F1 score of 0.7226, outperforming the naive 0.50 cutoff.

CHAPTER 5
SUMMARY, CONCLUSION, RECOMMENDATION

5.1 Summary

CreditRisk AI delivers an explainable, end-to-end machine learning system for credit delinquency prediction and borrower risk profiling. The project rigorously implements data validation, leakage-free preprocessing, class imbalance handling, multi-model benchmarking (Logistic Regression, Random Forest, XGBoost), decision threshold calibration, and SHAP game-theoretic explainability within a responsive Streamlit fintech dashboard.

5.2 Conclusion

The project demonstrates that combining gradient boosted decision trees with cost-sensitive threshold tuning significantly enhances minority delinquency detection compared to standard uncalibrated baselines. XGBoost achieved a state-of-the-art PR-AUC of 0.8654 and an F1 score of 0.7226. Furthermore, integrating SHAP attribution resolves the black-box dilemma, ensuring compliance with adverse action and fair lending requirements.

5.3 Recommendations

Future improvements include direct integration with credit bureau APIs (CIBIL/Experian), macroeconomic stress testing (interest rate shocks), and automated fair lending disparity auditing (ECOA / Reg B).

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